

Virtual Memory & File Systems - Quick Revision Cheat Sheet

QUICK REVISION & EXAM CHEAT SHEET

Subject: Operating Systems | Chapter: Chapter 3: Memory Management, Virtual Memory & File Systems | Academic Year: 2026-2027

Paging & TLB Calculations

- * Logical address: Page Number (p) + Offset (d). Number of pages = 2^p ; Page size = 2^d .
- * Effective Access Time (EAT) = $(\text{TLB_Hit_Ratio} * (\text{TLB_Time} + \text{Mem_Time})) + ((1 - \text{TLB_Hit_Ratio}) * (\text{TLB_Time} + 2 * \text{Mem_Time}))$.
- * Paging eliminates external fragmentation, but incurs internal fragmentation on the final page frame.

Page Replacement Algorithms Comparison

- * FIFO: Replaces oldest loaded page. Suffers from Belady's Anomaly.
- * LRU: Replaces page least recently referenced. Free from Belady's Anomaly.
- * Optimal: Replaces page that will not be used for longest time in future. Minimizes page faults.

Disk Scheduling Algorithms

- * SSTF selects request with minimum seek time from current head position; may starve distant requests.
- * SCAN sweeps from one end to the other servicing requests; reverses direction at disk boundary.
- * C-SCAN sweeps in one direction servicing requests, then immediately returns to start without servicing.